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Studierichting: Informatica
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$$I = \int \frac{4dx}{x^2 + x + 1} dx = \int \frac{4dx}{x^2 + x\frac{1}{4} + \frac{3}{4}} dx = \int \frac{4dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx$$

$$\text{Stel: } t = x + \frac{1}{2} \Rightarrow dt = dx$$

$$= 4 \int \frac{1}{t^2 + \sqrt{\frac{3}{4}}} dt = 4 \cdot \frac{1}{\sqrt{\frac{3}{4}}} \text{Bgtan} \left(\frac{t}{\sqrt{\frac{3}{4}}} \right) + c$$

$$= \frac{8}{\sqrt{3}} \text{Bgtan} \left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + c = \frac{8\sqrt{3}}{3} \text{Bgtan} \left(\frac{(2x + 1)\sqrt{3}}{3} \right) + c$$

References